QRE2026

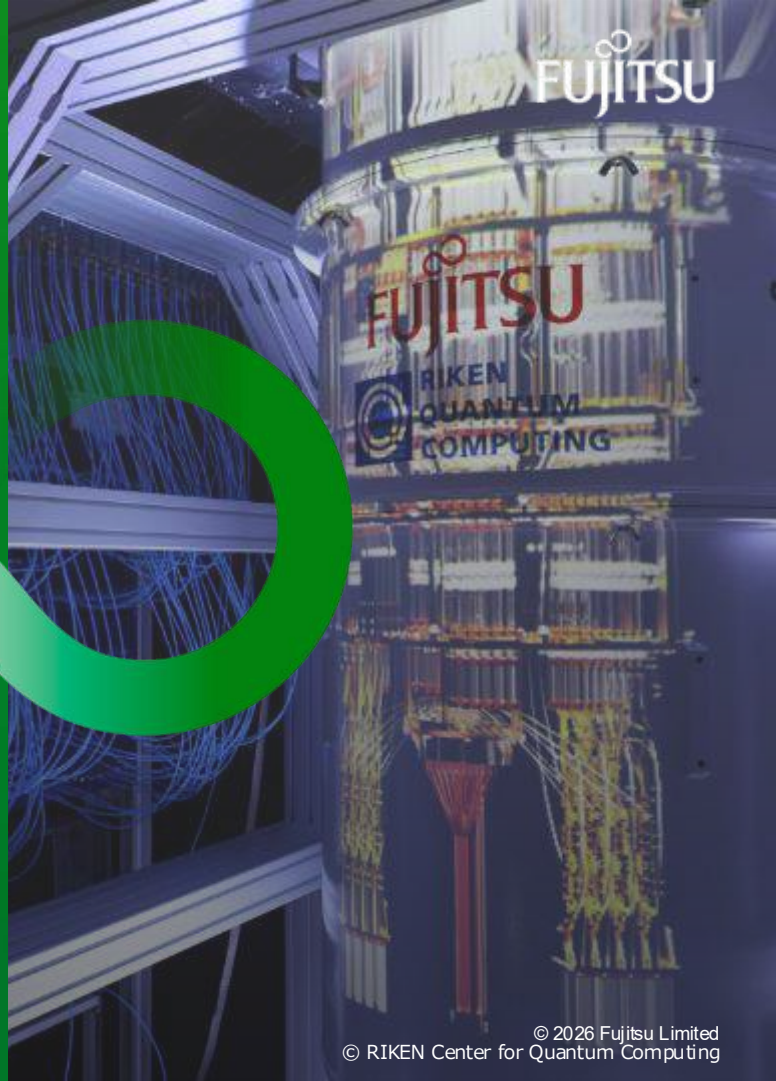
The 8-th International Workshop on Quantum Resource Estimation for Meaningful FT-Algorithms and Enabling QC Architectures

21 September 2026 @Toronto, Canada

Efficient Analog Rotations for Partially Fault-Tolerant Quantum Computing

Shota Kanasugi

Fujitsu Limited, Japan



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Outline

- Introduction
  - Space-Time efficient Analog Rotation (STAR) architecture for Early-FTQC devices
    - Y. Akahoshi et al., PRX Quantum **5**, 010337 (2024).
- STAR ver. 2
  - Improvement of the STAR architecture for *small-angle* rotations
    - R. Toshio et al., PRX 15, 021057 (2025).
    - Y. Akahoshi et al., PRX Quantum 6, 040319 (2025).
- STAR ver. 3
  - The latest STAR architecture based on Clifford+T+Φ gate set
    - R. Toshio, **SK** et al., arXiv:2603.22891 (2026).
- Resource estimation for STAR ver.3
  - Application to quantum chemistry
    - **SK**, R. Toshio et al., arXiv:2603.22891 (2026).

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Fujitsu's Quantum Computing Strategy

- Develop the full technology stack with world-leading research institutions.
- Focus on software while pursuing multiple hardware platforms.
- Co-develop applications with end users through our hybrid quantum computing platform.

Quantum Application	Research with end-user input: **Materials** | **Drug discovery** | **Finance**		**FUJIFILM, Tokyo Electron, etc.**	**TU Delft**
Quantum Software	Algorithm	**UC Berkeley** Quantum Error Correction	**U of Osaka** Quantum Error Correction	
Quantum Platform	Complier Middleware	**U of Edinburgh** Hybrid Computing	Cloud Technology	
Quantum State Control **Quantum Device & Integration**	**RIKEN** Superconducting Qubit	**TU Delft** Diamond Spin Qubit	**Yaqumo** Neutral Atom Qubit	

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Introduction

Space-Time efficient Analog Rotation (STAR) architecture for Early-FTQC devices

Y. Akahoshi et al., PRX Quantum **5**, 010337 (2024).

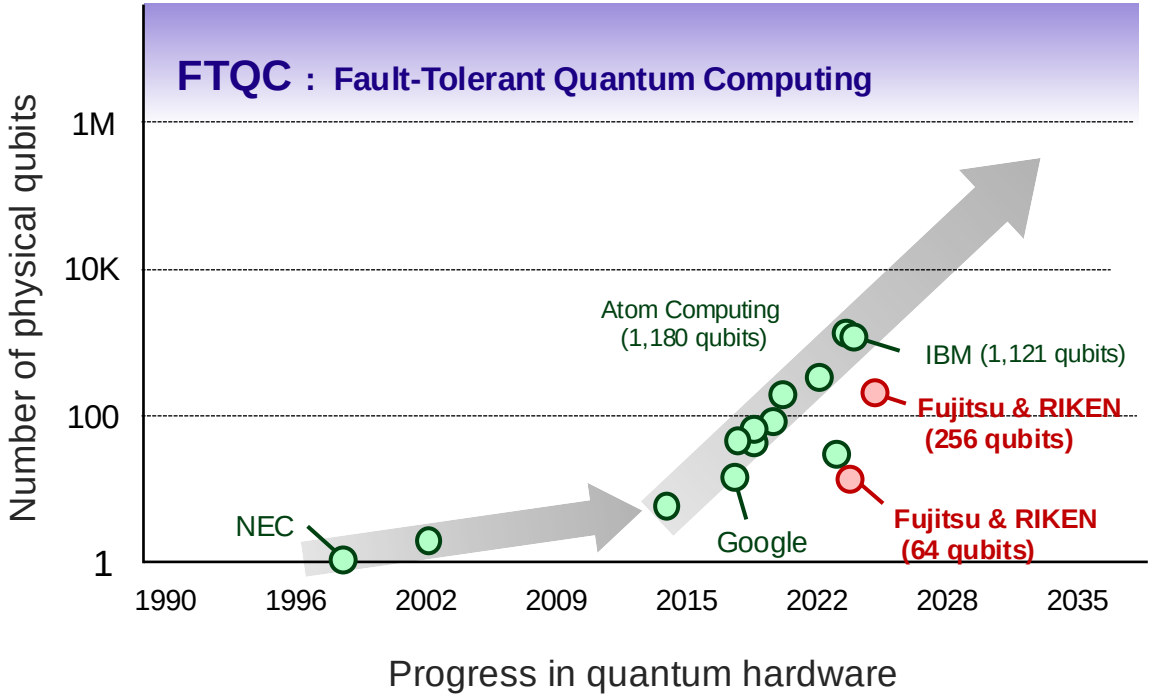
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Current state of quantum computers

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Practical FTQC applications remain far beyond the near-term horizon.



Under conventional FTQC, practical applications require millions of physical qubits.

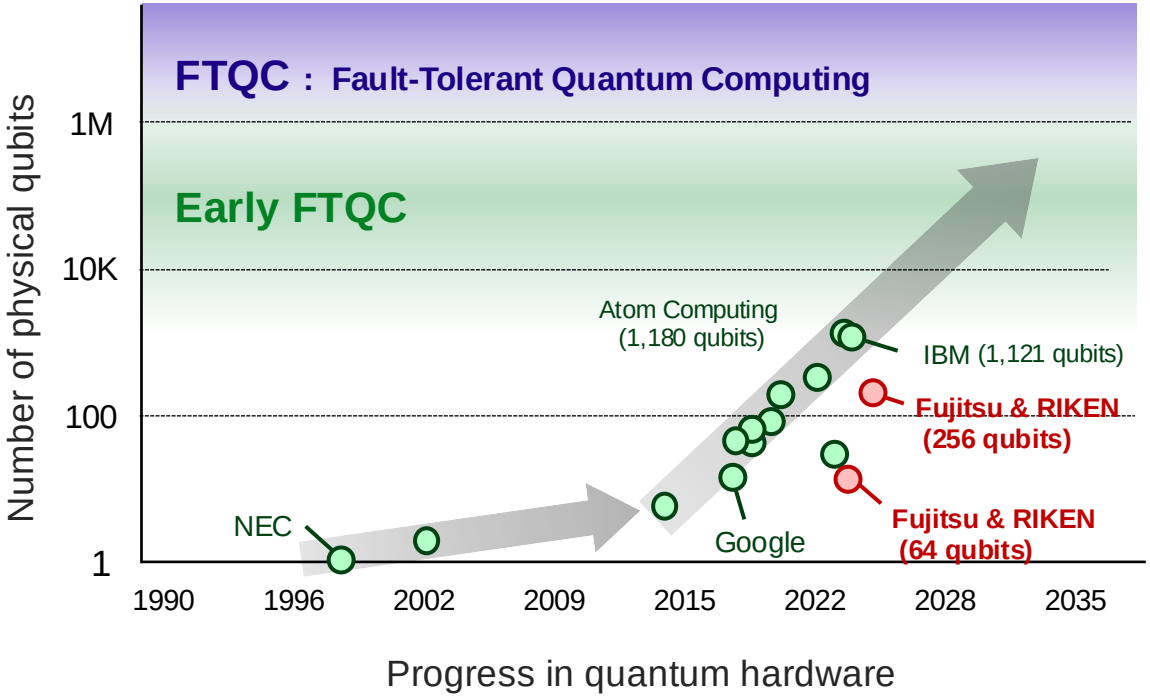
- **2048-bit Shor's algorithm:** 1×10^6 qubits (Gidney2025)
- **Quantum Chemistry**: 4×10^6 qubits (Low2025)
- **Condensed-matter physics**: $O(10^5)$ qubits (Yoshioka2022)

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Current state of quantum computers

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Practical FTQC applications remain far beyond the near-term horizon.

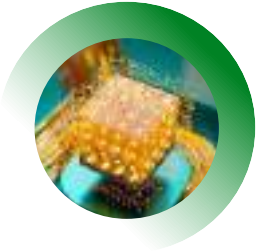


Under conventional FTQC, practical applications require millions of physical qubits.

A novel quantum computing architecture:

"STAR architecture"

(**S**pace-**T**ime efficient **A**nalog **R**otation)



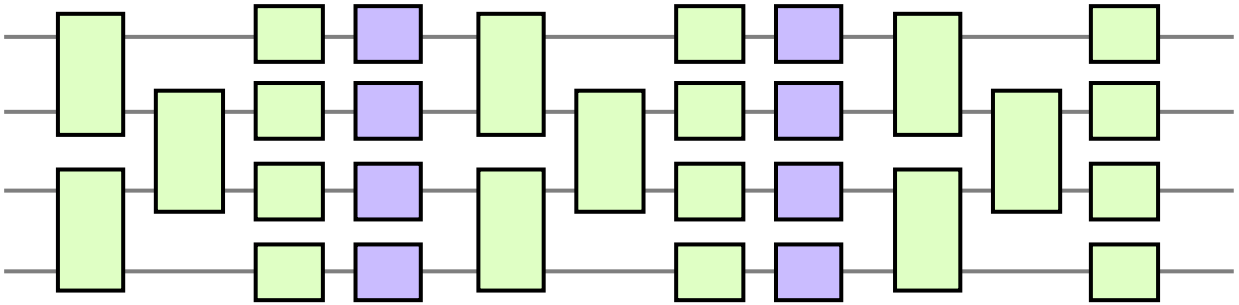
Y. Akahoshi et al., PRX Quantum **5**, 010337 (2024).

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Structure of practical quantum circuits

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- Practical quantum circuits typically use the Clifford+Φ gate set.
  - e.g., Trotterized/variational quantum circuits



Clifford gates

Analog Z-rotation gates $R_Z(\phi) := e^{i\phi Z}$

Efficiently implementable via lattice surgery with surface codes.

Compiled into Clifford+T gates
→ High space-time cost
→ Major bottleneck for FTQC

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Analog rotations in FTQC architectures

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- Implementing arbitrary rotations in FTQC requires two costly processes.

① Gate synthesis

N. J. Ross & P. Selinger, Quantum Information and Computation **16**, 901 (2016).

- Gate synthesis approximates $R_Z(\theta)$ with a sequence of $\{T, S, H\}$ gates.

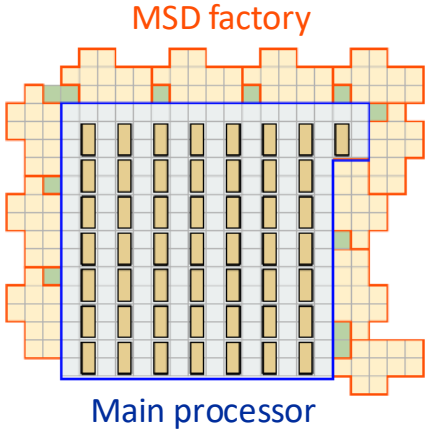
$$R_z(\pi/128) = \mathrm{HTSHTSHTSHTHTHTHTSHTHTSHTSHTSHTHTHTSHTSHTHTHTSHTHTSHTHTHTHTHTHTHTSHTSHT}$$
$$\mathrm{SHTHTSHTHTSHTHTHTHTSHTHTHTSHTHTSHTHTHTHTSHTSHTSHTHTHTSHTSHTSHTSHTHTSHTS}$$
$$\mathrm{HTSHTSHTHTSHTHTSHTSHTHTHTHTHTSHTHTHTHTSHTSHTSHTHTSHTSHTHTHTSHTHTHTHTHTS}$$
$$\mathrm{HTSHTHTHTHTHTSHTHTHTHTSHTHTHTHTHTHTHTH}\omega^7$$

→ Many T-gates; **large time-overhead**

② Magic state distillation (MSD)

- Each T-gate consumes a high-fidelity magic state produced via MSD.

→ Multiple MSD factories are needed for a steady magic-state supply, resulting in **large spatial overhead.**



D. Litinski, Quantum 3, 205 (2019).

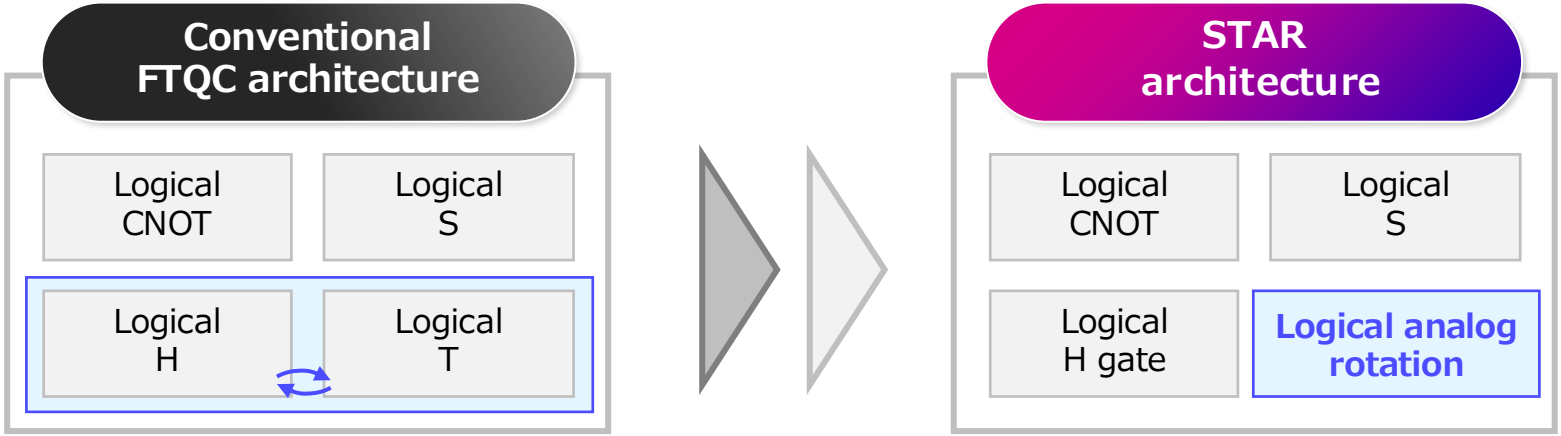
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Core idea of the STAR architecture

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- STAR replaces the conventional Clifford+T gate set with **Clifford+Φ**.

Y. Akahoshi et al., PRX Quantum **5**, 010337 (2024).

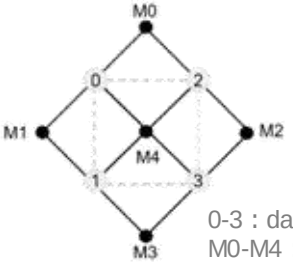


Synthesized rotation

➢ Trade-off: Analog rotations remain noisy, so STAR is only **partially fault tolerant**.

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Analog rotations in STAR



0-3 : data qubit
M0-M4 : measurement qubit

- Analog rotations are implemented directly using ***resource-state preparation*** and ***gate teleportation***.

① Resource state preparation

The resource state $|m_\theta\rangle_L$ is prepared *non-fault-tolerantly* using [[4,1,1,2]] code.

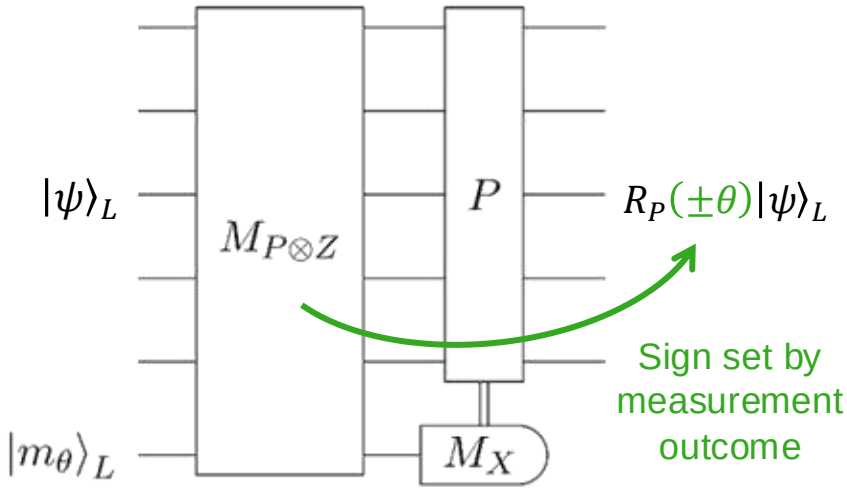
$$|m_\theta\rangle_L \equiv R_z(\theta)|+\rangle_L = \cos\theta\,|+\rangle_L + i\sin\theta\,|-\rangle_L$$

② Gate teleportation

We implement an analog rotation by injecting $|m_\theta\rangle_L$ into the circuit shown on the right.

※ The inverse rotation occurs with probability 1/2.

Gate teleportation circuit

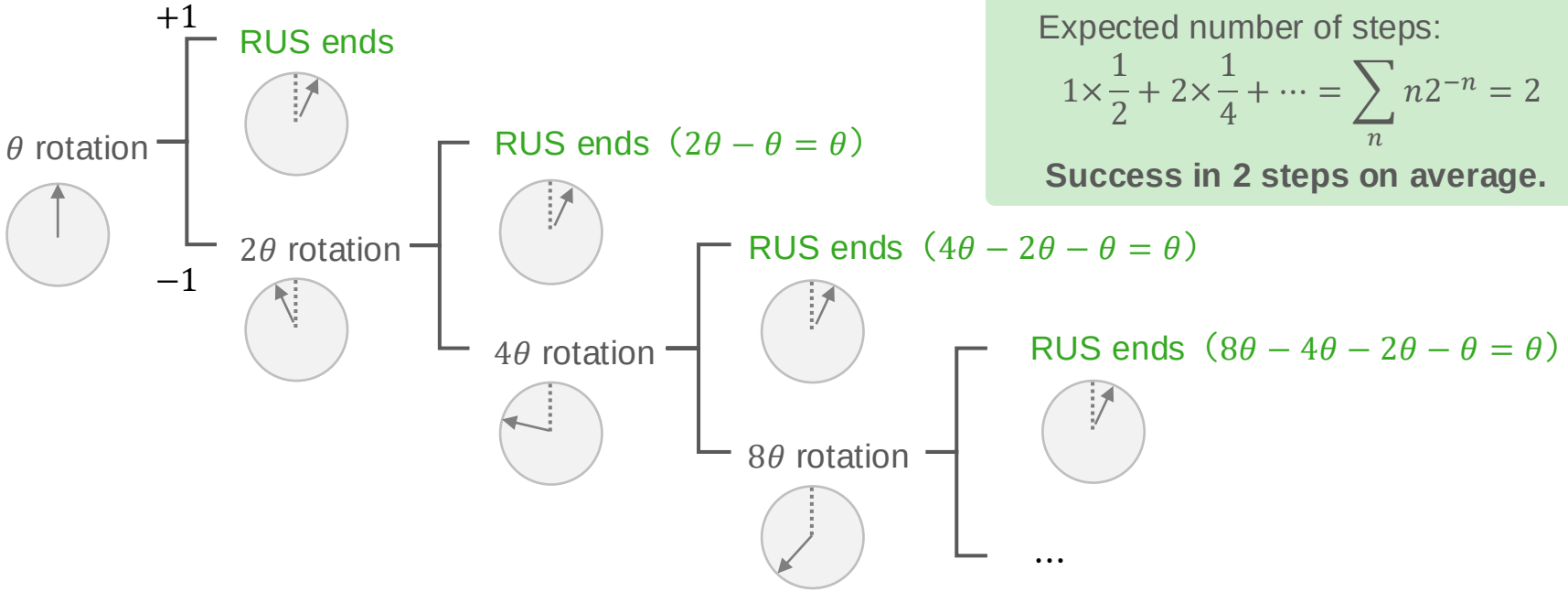


P : an arbitrary Pauli string (e.g., XXZ)

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Analog rotations in STAR

- A failed teleportation is corrected by retrying with twice the rotation angle.



Expected number of steps:

$$1\times\frac{1}{2}+2\times\frac{1}{4}+\cdots=\sum_n n2^{-n}=2$$

Success in 2 steps on average.

- Repeated until the target rotation is obtained (**Repeat-Until-Success; RUS**)

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Evolution of the STAR architecture

- STAR admits several variants depending on the **resource-state preparation** and **RUS protocol**.
- STAR has evolved significantly from v1 to v3, **greatly improving analog-rotation accuracy.**

Improved scaling with rotation angle θ_L

Architecture	Gateset	T-gate	Analog Rotation	Scaling of $P_L(\theta_L)$
FTQC [14, 31]	Clifford+T or Clifford+Toffoli	MSD or/and MSC	Gate synthesis	—
STAR ver. 1 [46]	Clifford+ϕ	—	[[4,1,1,2]] protocol	$\mathcal{O}(p_{\text{ph}})$
STAR ver. 2 [48]	Clifford+ϕ	—	TMR protocol and [[4,1,1,2]] protocol	$\mathcal{O}(\theta_L p_{\text{ph}})$
STAR ver. 3 (this work)	Clifford+T+ϕ	MSD or/and MSC	STAR-magic mutation	$\mathcal{O}(\theta_L^{2(1-\Theta(1/d))} p_{\text{ph}})$

STAR v3 also incorporates T gates.

Improved resource-state preparation protocols

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STAR ver.2

Improvement of the STAR architecture for *small-angle* rotations.

R. Toshio et al., PRX **15**, 021057 (2025).
Y. Akahoshi et al., PRX Quantum **6**, 040319 (2025).

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Transversal Multi-Rotation (TMR) protocol

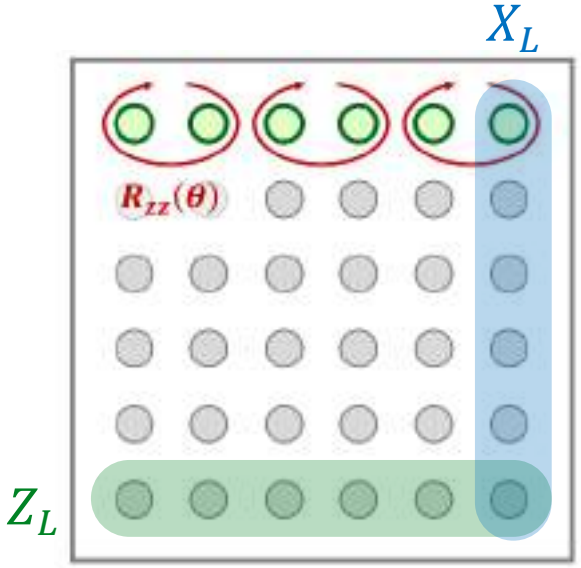
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- TMR enables more accurate resource-state preparation for *small-angle rotations*.

A Transversal multi-Pauli rotation yields a logical rotation plus effective Pauli errors.

Choi et al., arXiv:2303.17380 (2023) ; Toshio et al., PRX **15**, 021057 (2025).

$$\prod R_{ZZ}(\theta) = \cos^3\theta \cdot IIIIII = I_L$$
$$+i\cos^2\theta\sin\theta \cdot (ZZIIII + IIZZII + IIIIZZ)$$
$$-\cos\theta\sin^2\theta \cdot (ZZZZII + ZZIIZZ + IIZZZZ)$$
$$-i\sin^3\theta \cdot ZZZZZZ = Z_L$$
$$= \cos^3\theta \cdot I_L - i\sin^3\theta \cdot Z_L + \text{(Pauli error terms)}$$
$$\propto R_{L,Z}(\theta_*)$$
$$\theta_* \equiv -\arcsin\left(\frac{\sin^3\theta}{\sqrt{\cos^6\theta+\sin^6\theta}}\right) \simeq \theta^3$$



Ex. distance-6 surface code

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Transversal Multi-Rotation (TMR) protocol

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- TMR enables more accurate resource-state preparation for *small-angle rotations*.

Apply a transversal rotation to $|+\rangle_L$, then measures the stabilizers.

STEP①: Apply a transversal multi-Pauli rotation on $|+\rangle_L$

$$\prod R_{ZZ}(\theta)\,|+\rangle_L \propto R_{L,z}(\theta_*)\,|+\rangle_L + (\text{Pauli error})\,|+\rangle_L$$

- Target resource state
- All stabilizer eigenvalues: $+1$

- Outside the code space
- At least one stabilizer eigenvalue: -1

STEP②: Measure the stabilizes twice

Post-select runs with no error syndrome:

$$\prod R_{ZZ}(\theta)\,|+\rangle_L \xrightarrow{post-selection} |m_{\theta_*}\rangle_L = R_{L,z}(\theta_*)\,|+\rangle_L$$

Here, we neglect physical errors.

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Noise in the TMR protocol

- With physical noise, the output contains an erroneous resource-state component with weight $\mathcal{O}(\theta_*^{2/k} p_{\mathrm{ph}})$.

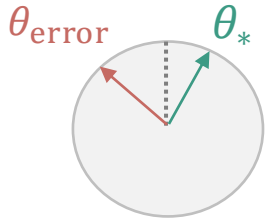
  The post-selected output state is:

$$\hat{\rho}_{out} \simeq (1-P)\cdot \left|m_{\theta_*}\right\rangle\!\left\langle m_{\theta_*}\right|_L + P\cdot \left|m_{\theta_{error}}\right\rangle\!\left\langle m_{\theta_{error}}\right|_L + \mathcal{O}(p_{\mathrm{ph}}^2)$$

$$= \mathcal{O}(\theta_*^{2/k} p_{\mathrm{ph}}) \qquad \propto -\theta_*^{1-2/k}$$

Target resource state

Erroneous state with an incorrect angle



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Noise in the TMR protocol

- With physical noise, the output contains an erroneous resource-state component with weight $\mathcal{O}(\theta_*^{2/k} p_{\mathrm{ph}})$.

  The post-selected output state is:

$$\hat{\rho}_{out} \simeq (1-P)\cdot |m_{\theta_*}\rangle\langle m_{\theta_*}|_L + \underbrace{P}_{=\mathcal{O}(\theta_*^{2/k} p_{\mathrm{ph}})} \cdot \underbrace{|m_{\theta_{error}}\rangle\langle m_{\theta_{error}}|_L}_{\propto -\theta_*^{1-2/k}} + \mathcal{O}(p_{\mathrm{ph}}^2)$$

Remark The erroneous rotation $R_{z,L}(\theta_{error})$ remain close to the target $R_{z,L}(\theta_*)$, limiting its impact:

Error magnitude ≈ $\left\| R_{z,L}(\theta_* - \theta_{error}) - I \right\| \propto \theta_*^{1-2/k}$

➡ Effective error rate: $P_{\mathrm{eff}} \simeq P \times \theta_*^{1-2/k} \simeq \mathcal{O}(\theta_* p_{\mathrm{ph}})$

→ Especially advantageous for *small-angle* rotations ($\theta_* \ll 1$)

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STAR ver.3

The latest STAR architecture based on Clifford+T+Φ gate set

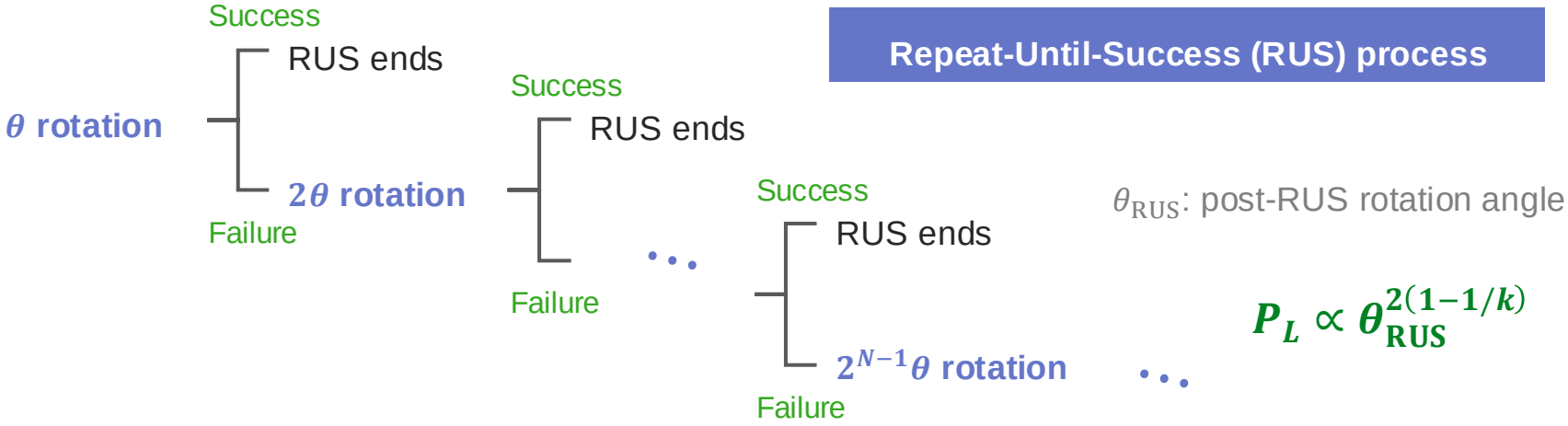
R. Toshio, **SK** et al., arXiv:2603.22891 (2026).

(In collaboration with The University of Osaka)

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Challenge | Bottleneck in the TMR Protocol

- Background: Limitation of STAR ver. 2
  - Long RUS tails dominate the average error → mean accumulated error $\mathcal{O}(\theta_L p_{ph})$



Each failure doubles the angle → errors grow quadratically and accumulate.

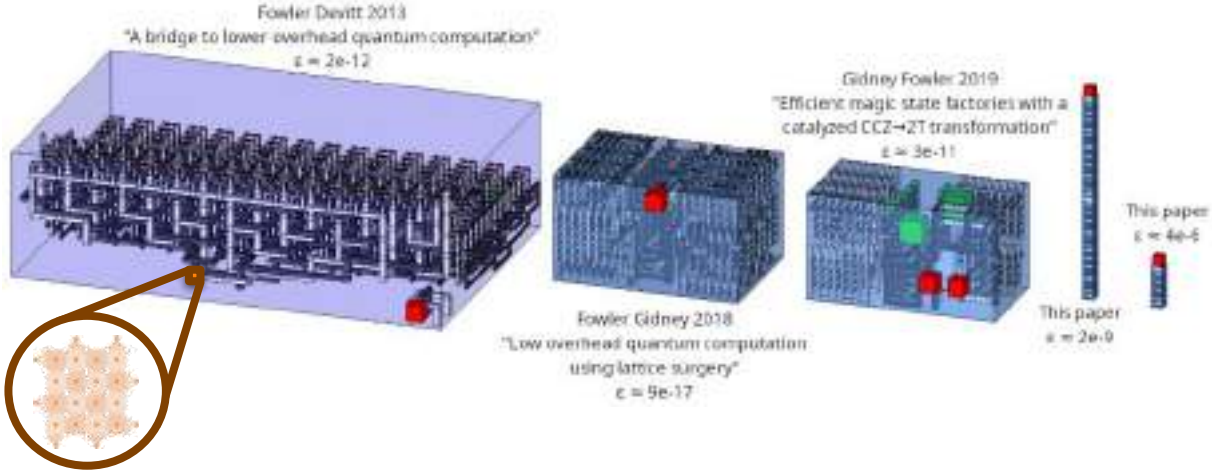
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Background | Magic-State Cultivation (MSC)

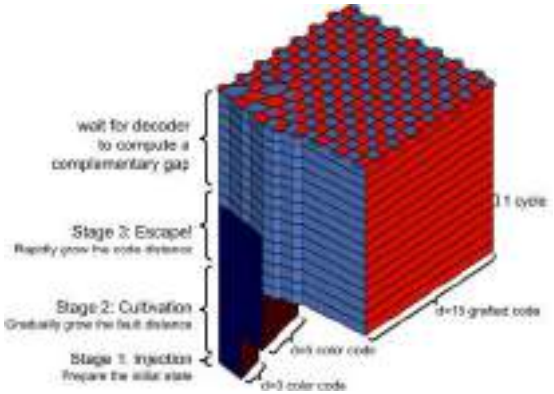
- Google introduced a breakthrough method that cuts T-gate costs.

C. Gidney et al., arXiv:2409.17595

✓ **Drastically reduced space-time cost of magic state factory.**



✓ **Single-patch preparation**



Both STAR and MSC can be executed in a single surface-code patch.

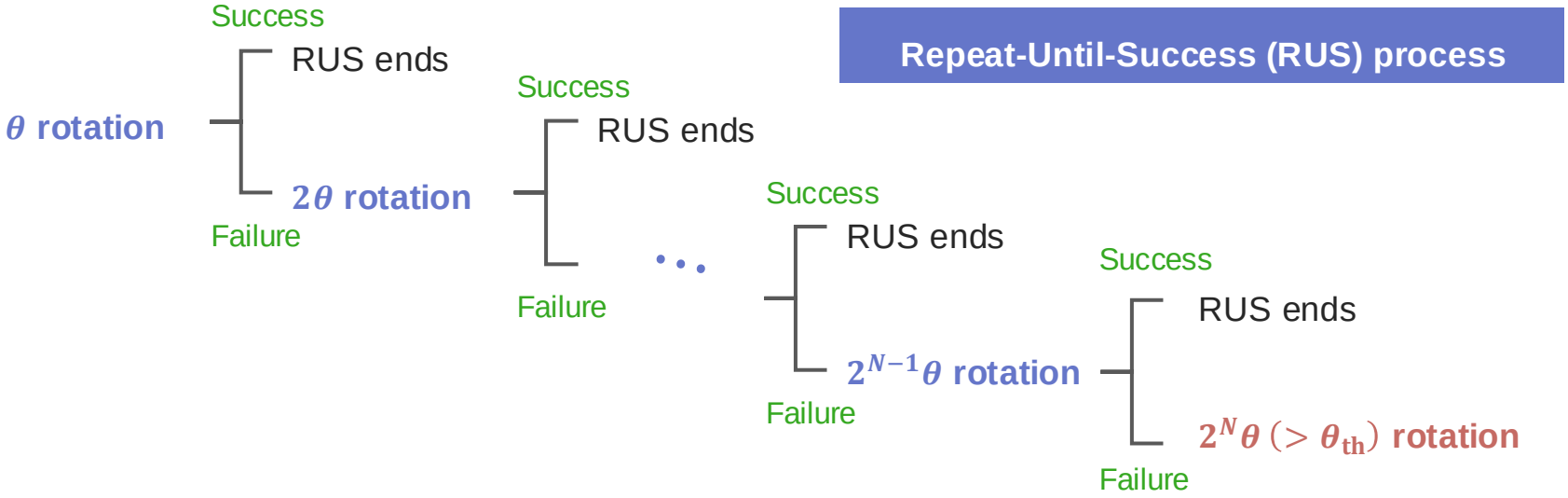
→ What can we do if we combine these approaches?

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Proposal | STAR-Magic Mutation (STAR ver.3)

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- Solution: STAR-magic mutation
  - Once RUS attempts exceed a threshold, switch to cultivation-based T synthesis.



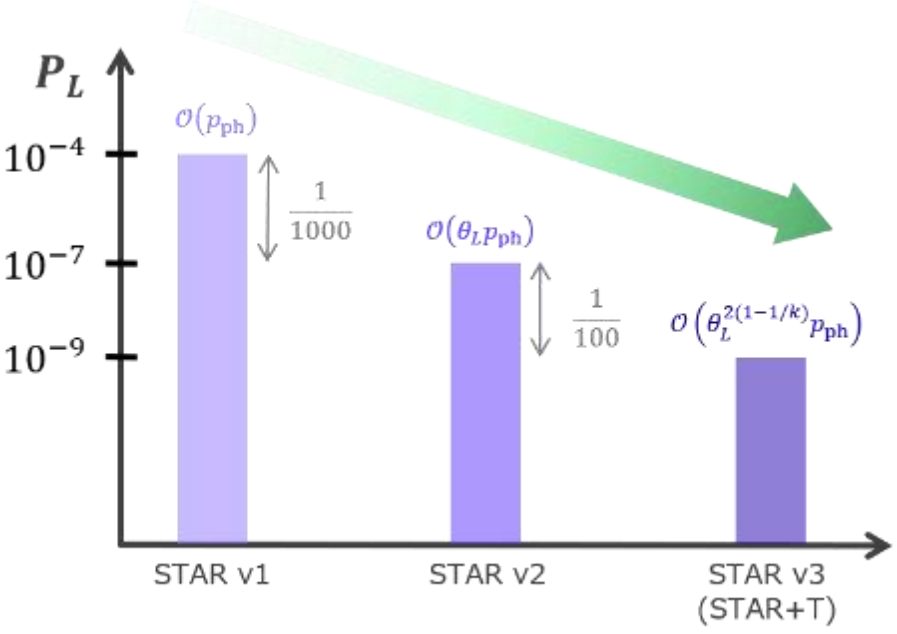
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Provable improvement of logical error rate

- STAR-magic mutation improves the error scaling of logical $R_Z(\theta_L)$:

$$\mathcal{O}(\theta_L p_{ph}) \rightarrow \mathcal{O}(\theta_L^{2(1-1/k)} p_{ph})$$

※ Valid when logical T-gate errors are negligible.



Theorem 1 (Asymptotic behavior of STAR-magic mutation under the supply of perfect magic states)**.** *Let us assume an ideal case where we can supply perfect magic states in the digital stage. In this case, the STAR-magic mutation achieves the worst-case error rate of an analog rotation gate $R_{z,L}(\theta_L)$ with the following scaling:*

$$\tilde{P}_L(\theta_L) \simeq \mathcal{O}(\theta_L^{2(1-1/k)} p_{\mathrm{ph}}) \qquad (r : \text{fixed};\ \theta_L \to 0)$$

$$\tilde{P}_L(\theta_L) \simeq \mathcal{O}(\theta_L p_{\mathrm{ph}}) \qquad (\theta_{\mathrm{th}} : \text{fixed};\ \theta_L \to 0)$$

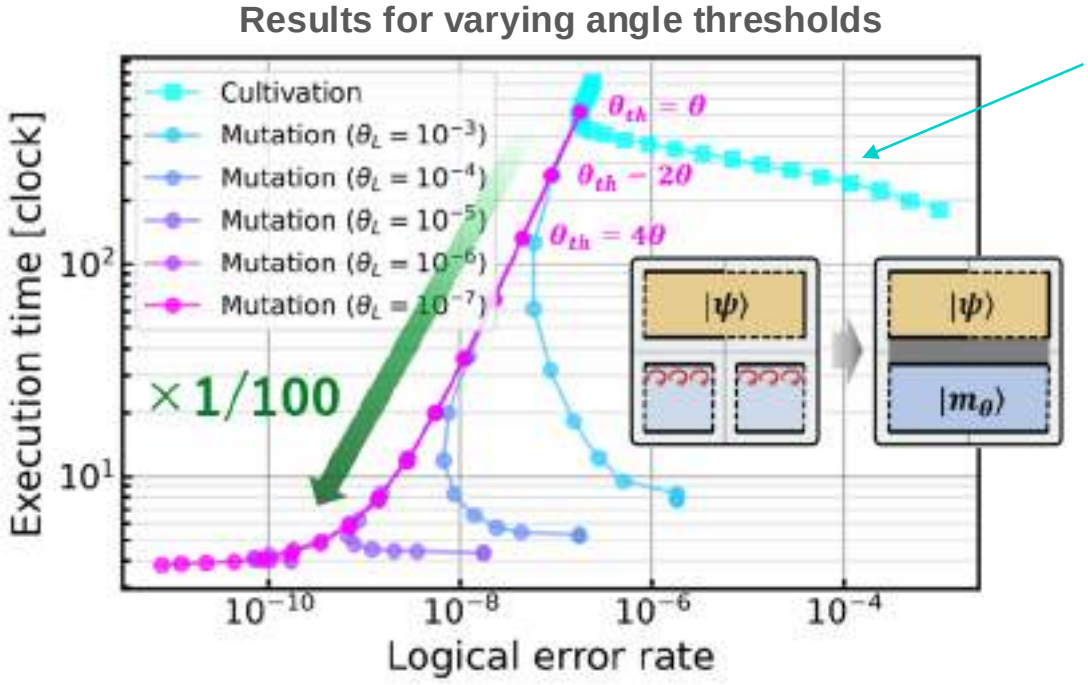
where $r = \theta_{\mathrm{th}}/\theta_L$ is the ratio between the threshold angle θ_{th} and the target angle θ_L.

R Toshio, **SK** et al., arXiv:2603.22891 (2026).

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Numerics | Runtime-accuracy trade-off

- STAR-magic mutation is **100× faster** and **100×-higher precision** than cultivation-based T-gate synthesis for $\theta_L \lesssim 10^{-5}$.



Synthesis error is varied for MSC

- $p_{\mathrm{ph}} = 10^{-3}$
- T-gate error rate (MSC): 2×10^{-9}
- 10 clock / magic state
- two patches for TMR / cultivation
- threshold: $\theta_{\mathrm{th}} = 2^n \theta_L$

R Toshio, **SK** et al., arXiv:2603.22891 (2026).

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Resource estimation for STAR ver.3

Application to quantum chemistry

SK, R. Toshio et al., arXiv:2603.22891 (2026).

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A target application for STAR

- **Trotter simulation** is a promising application for STAR.

Hamiltonian

$$\hat{\mathcal{H}} = \sum_{i=1}^{L} a_i \hat{P}_i$$

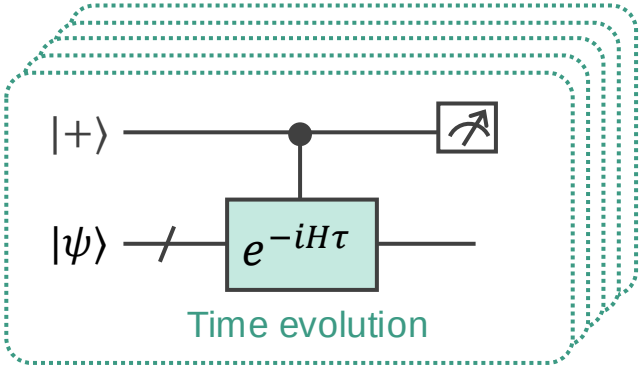
Trotter circuit

$$e^{-iT\hat{\mathcal{H}}} \simeq \left(\prod_{i=1}^{L} e^{-i\left(\frac{a_i T}{2N}\right)\hat{P}_i} \prod_{i=L}^{1} e^{-i\left(\frac{a_i T}{2N}\right)\hat{P}_i} \right)^N$$

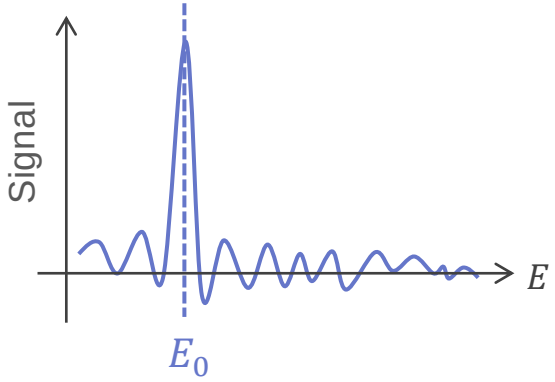
Typically, each rotation angle is below 10^{-3}.

- We consider **single-ancilla phase estimation** using Trotter circuits.

① Repeat the Hadamard test at many evolution times τ.



② Extract phase information classically from the measurement outcomes.



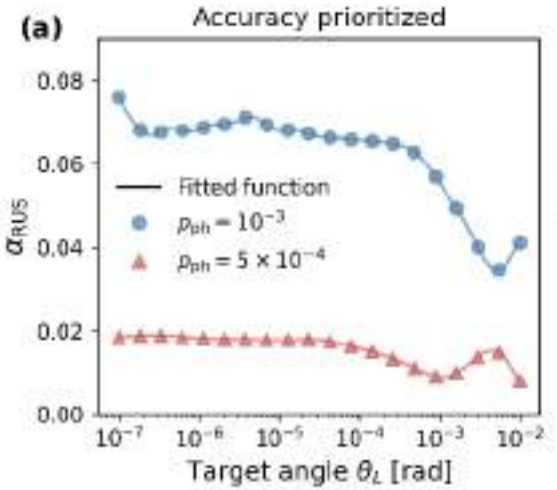
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Circuit design for STAR

- Each logical rotation in STAR has logical error rate

$$P(\theta_L) \simeq \alpha_{\mathrm{RUS}}(\theta_L)\theta_L p_{\mathrm{ph}}$$

α_{RUS} captures RUS effects and depends weakly on θ_L.



- The total accumulated error from non-Clifford rotations with angles $\{\theta_i\}$ is

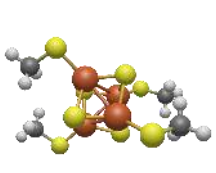
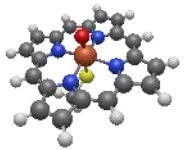
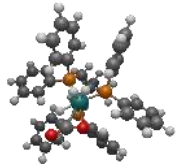
$$P_{\mathrm{total}} \equiv \left(\sum_i \alpha_{\mathrm{RUS}}(\theta_i)\, \theta_i \right) p_{\mathrm{ph}}$$

→ Probabilistic error cancellation mitigates this error with sampling overhead $\gamma^2 = e^{4P_{total}}$.

To prevent runtime from growing exponentially, minimizing P_{total} is crucial.

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Resource estimation setup

- Problem
  - Ground-state energy estimation with chemical accuracy.
  - Active-space models for Fe-S clusters, cytochrome P450, and CO_2-utilization catalysts.
- Algorithmic setup
  - Single-ancilla phase estimation (robust phase estimation) [Ni et al., Quantum 7, 1165 (2023)]
  - Time evolution: partially randomized Trotterization [Günther et al., PRXQ 7, 020332 (2026)]
  - Hamiltonian optimization: **Unitary Weight Concentration (UWC*)** [SK et al., arXiv:2603.22891 (2026)]

    *Hamiltonian optimization that reduces rotation counts in partially randomized Trotter circuits

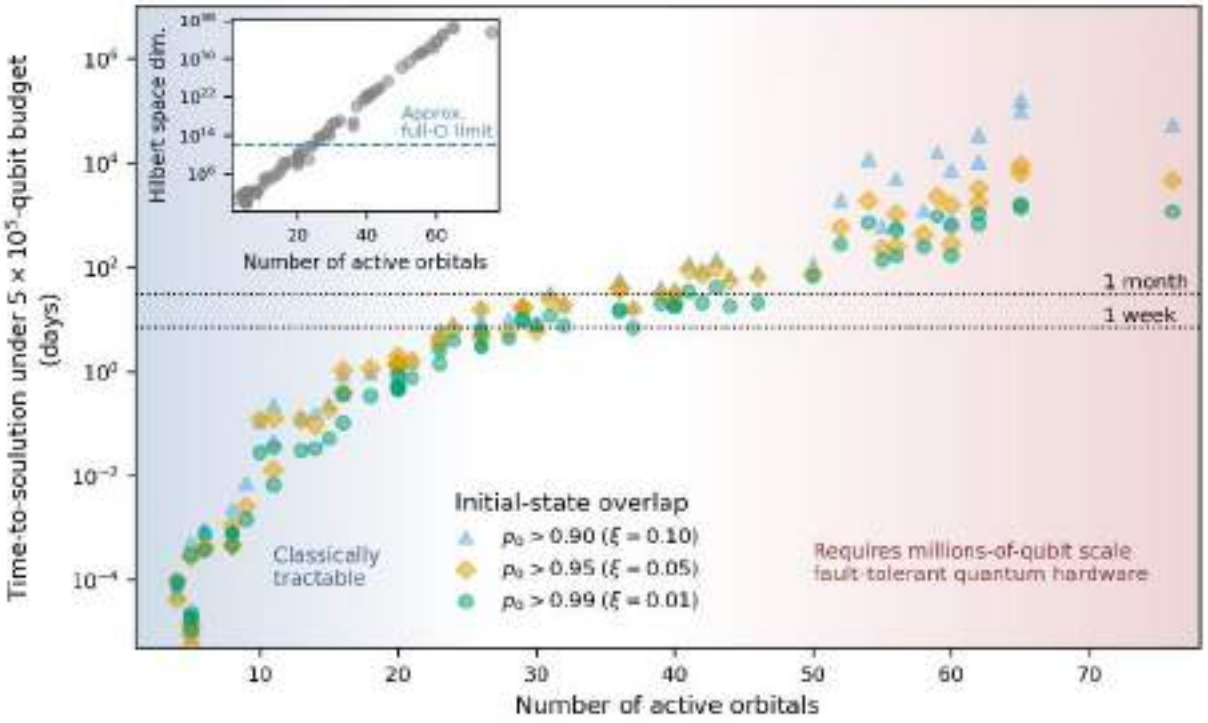
    → **Partial randomization + UWC reduces gate count and runtime by up to ~1000x versus standard second-order Trotterization.**
- Architecture setup
  - Surface code: fast block layout [Litinski, Quantum 3, 128 (2019)]
  - Optimized accuracy-speed trade-off for STAR-magic mutation



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Resource estimation results

- Feasibility map: Runtime-based target regime for early-FTQC devices

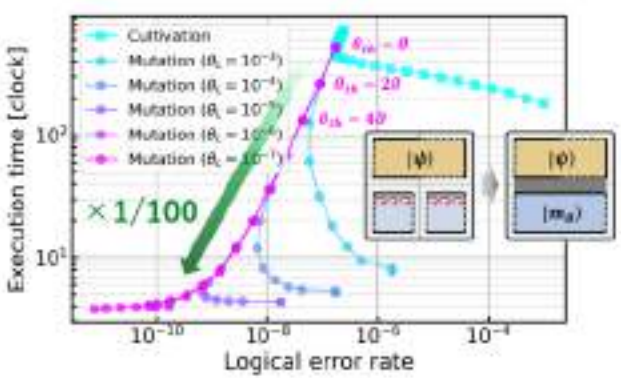


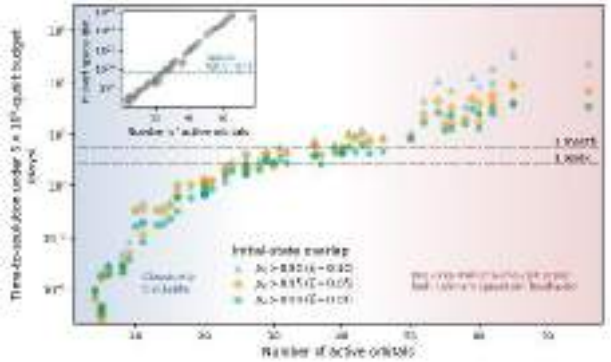
- ✓ Models with 20-50 active orbitals are feasible within weeks to months, beyond the reach of exact classical diagonalization.
- ✓ Models with more than 50 active orbitals remain out of reach due to exponentially growing error-mitigation overhead.
- ✓ State-preparation cost and runtime remain key challenges.

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Summary

- STAR is a **partially fault-tolerant architecture** for early-FTQC devices.
  - Error-corrected Clifford gates + noisy non-Clifford rotations
- STAR has evolved through three generations, progressively improving analog-rotation accuracy.
  - STAR v1: General-purpose analog rotations
  - STAR v2: High-precision small-angle rotations
  - STAR v3: Small-angle analog + T-synthesized rotations
    → **Achieves the precision needed for quantum chemistry simulations**
- Outlook
  - Demonstrate STAR on quantum hardware
  - Identify promising applications for STAR
  - Advance quantum error correction





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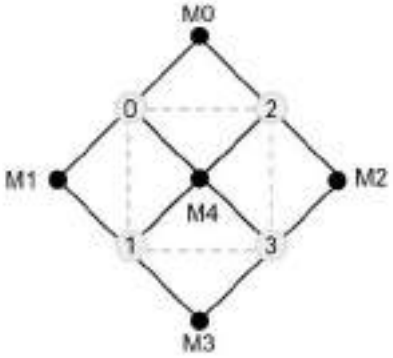
Thank you



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[[4,1,1,2]] code for state injection

- We employ the [[4,1,1,2]] code for the state injection
  - Encoding two logical degrees of freedom (DOF) using four physical qubits.



0-3 : data qubit
M0-M4 : measurement qubit

$$S_Z = Z_0Z_1Z_2Z_3,$$
$$S_X = X_0X_1X_2X_3$$

Logical DOF A → **ancilla state $|m_\theta\rangle_L$**

Logical DOF B → **gauge DOF**

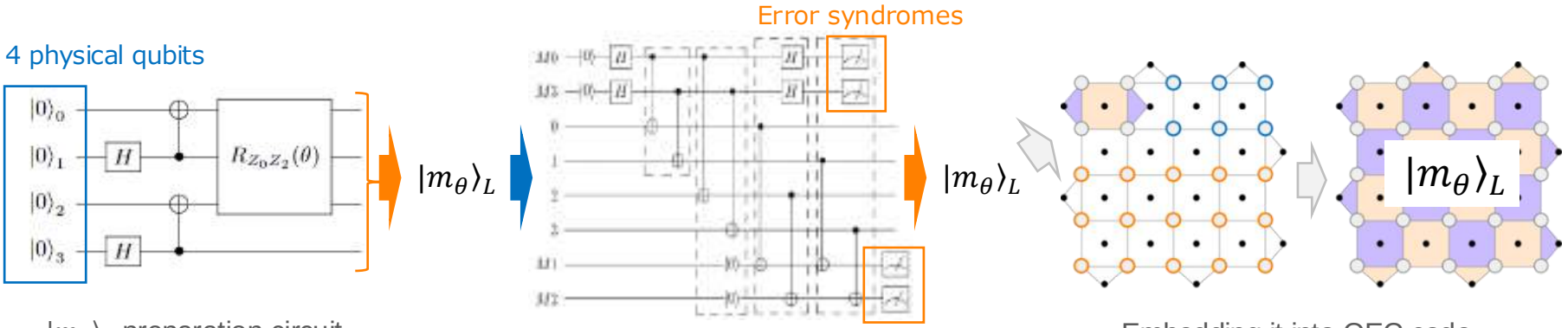
- ✓ Code distance is two => single errors can be detected
- ✓ **By utilizing the gauge DOF, measurement circuit can be fault-tolerant and shallow**

ex) $S_Z = (Z_0Z_1)(Z_2Z_3)$ (product of gauge operators)

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State injection protocol

- The ancilla state is prepared by post-selection



$|m_\theta\rangle_L$ preparation circuit ([[4,1,1,2]] code)

Error detection circuit

Embedding it into QEC code

Under the circuit-level noise (every operations suffer from errors with probability p), the error rate of the ancilla state behaves $p_L = \frac{2p}{15} + O(p^2)$

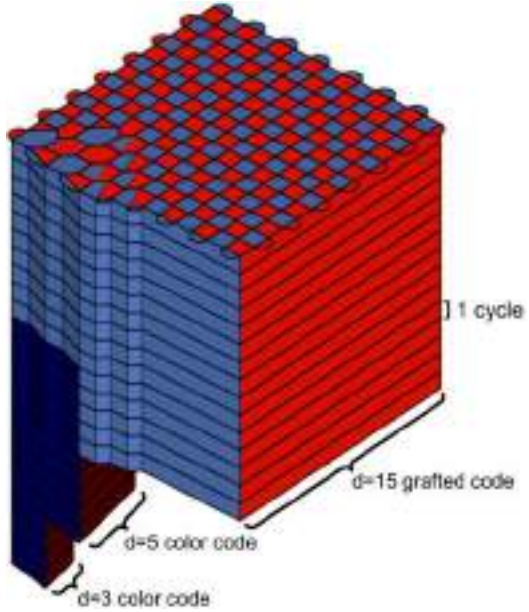
more accurate than previous study for magic state (e.g. $46p/15$ in Y. Li New J. Phys. 17, 023037 (2015))

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Magic-State Cultivation

- Google introduced a breakthrough method that cuts T-gate costs.

Single-patch preparation



High-fidelity T gates

- Low success probability
- Nonzero error floor

Paper	Noise Strength	Fault Distance	Error Rate (End to End)	Discard Rate (End to End)
Li 2015 [Li15]	10^{-3}	7	$5 \cdot 10^{-3}$	50%
Chamberland et al 2020 [CN20]	$5 \cdot 10^{-4}$	7	N/A	N/A
Chamberland et al 2020 [CN20]	10^{-4}	7	N/A	N/A
Gidney 2023 [Gid23]	10^{-3}	7	$6 \cdot 10^{-4}$	75%
Itogawa et al 2024 [Ito+24]	10^{-3}	3	N/A	N/A
(This Paper)	10^{-3}	3	$3 \cdot 10^{-6}$	80%
(This Paper)	10^{-3}	5	$2 \cdot 10^{-9}$	99%
(This Paper)	$5 \cdot 10^{-4}$	5	$4 \cdot 10^{-11}$	90%
(This Paper)	10^{-4}	5	N/A	N/A

C. Gidney et al., arXiv:2409.17595

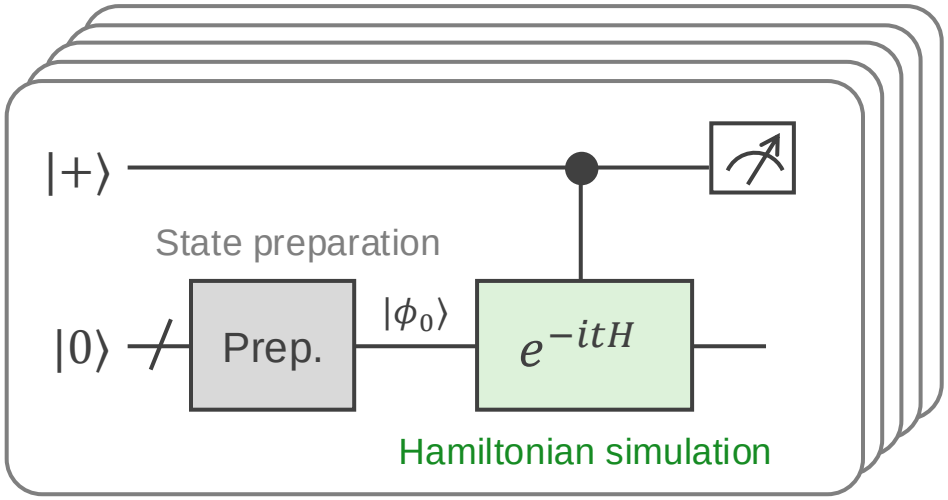
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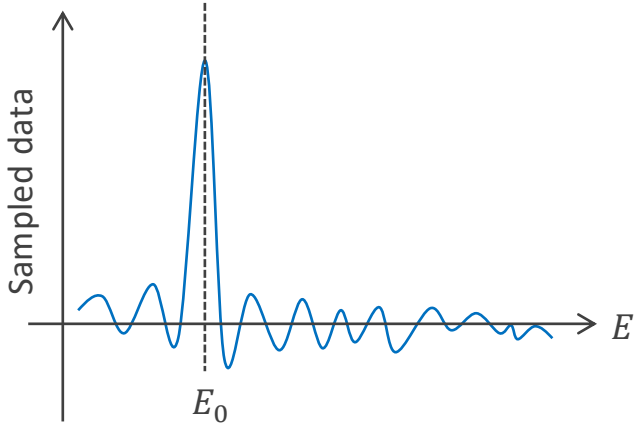
Single-ancilla phase estimation

- Quantum phase estimation for early-FTQC

Hadamard test sampling (QPU)

Classical postprocessing (CPU)





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Single-ancilla phase estimation

- We employ **Robust Phase Estimation (RPE)** algorithm. Ni et al., Quantum 7, 1165 (2023).
  - Achieving the Heisenberg limit for total runtime T_{total} (for exact time evolution).

    $T_{\mathrm{total}} = \tilde{\mathcal{O}}(\epsilon^{-1})$ ϵ: target accuracy
  - Maximum circuit depth T_{max} can be smaller for a high-fidelity input state $|\phi\rangle$.

$$T_{\mathrm{max}} = \tilde{\mathcal{O}}(\xi\epsilon^{-1}), \quad T_{\mathrm{total}} = \tilde{\mathcal{O}}(\xi^{-1}\epsilon^{-1}),$$
$$\xi \simeq 1 - p_0, \quad p_0 = |\langle\phi|\psi_0\rangle|^2$$

Algorithm 2 Phase estimation in the regime of large overlap

Input: ϵ: target accuracy, η: upper bound of the failure probability, δ: upper bound for the noise in the initial state $|\psi\rangle$, ξ: additional prefactor for the maximal runtime that must satisfy $1 > \xi > \frac{1}{\pi}\arcsin((1-\delta)^{-1}\epsilon)$.

Let $J = \lceil \log_2(\frac{\pi}{\epsilon}) \rceil$ and calculate N_s according to the values of the inputs and (16).

$\theta_{-1} = 0$.

for $j = 0, 1, \ldots, J$ **do**

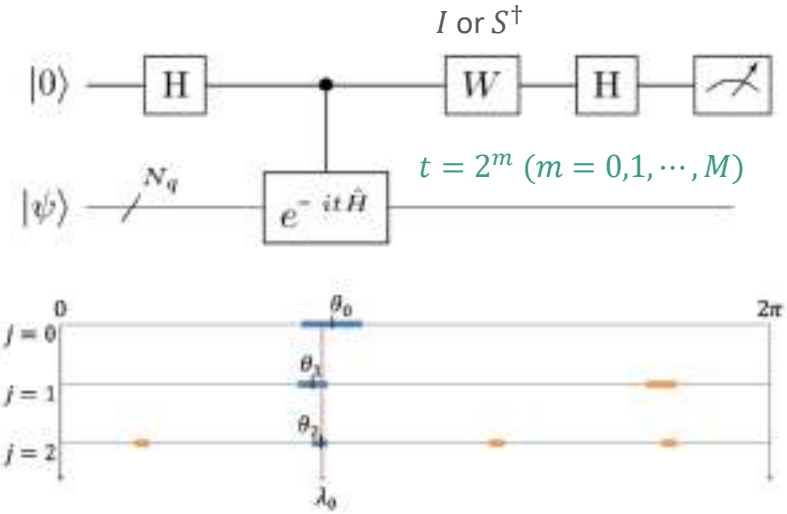
Run the circuit in Figure 1(b) for the real part and imaginary part of $\langle\psi|U^{2^j}|\psi\rangle$ for $\frac{N_s}{2}$ times each to generate Z_j as an estimation of $\langle\psi|U^{2^j}|\psi\rangle$.

Define a candidate set $S_j = \left\{\frac{2k\pi + \arg Z_j}{2^j}\right\}_{k=0,\ldots,2^j-1}$

$\theta_j = \arg\min_{\theta \in S_j} |\theta - \theta_{j-1}|_{2\pi}$

end for

Output: θ_J as an approximation to λ_0.



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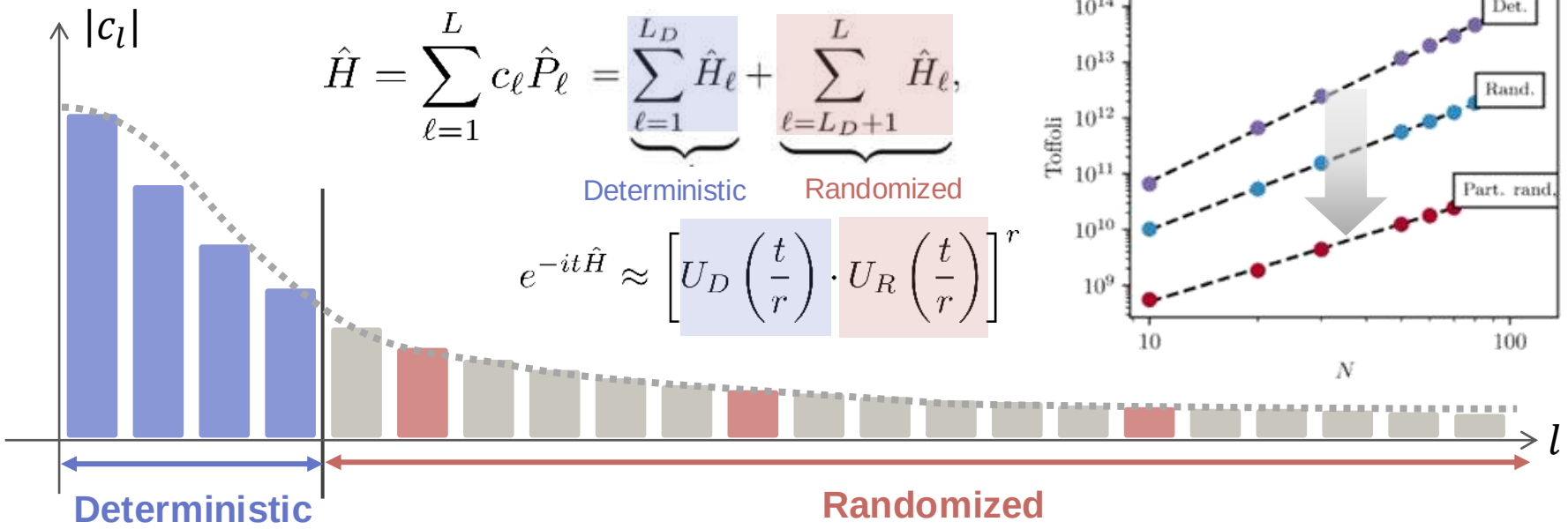
Partially randomized Trotterization

- The Pauli coefficients of molecular Hamiltonians are typically highly nonuniform.
  → Treating the large-weight terms **deterministically** and the remaining terms **randomly** enables more efficient Hamiltonian simulation [1].

[1] Günther et al., PRX Quantum 7, 020332 (2026).

$$\hat{H} = \sum_{\ell=1}^{L} c_\ell \hat{P}_\ell = \underbrace{\sum_{\ell=1}^{L_D} \hat{H}_\ell}_{\text{Deterministic}} + \underbrace{\sum_{\ell=L_D+1}^{L} \hat{H}_\ell}_{\text{Randomized}},$$

$$e^{-it\hat{H}} \approx \left[U_D\left(\frac{t}{r}\right) \cdot U_R\left(\frac{t}{r}\right) \right]^r$$



Result for hydrogen chain [1]

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Partially randomized Trotterization

- Comparison of cost (for RPE algorithm)

ϵ: target accuracy
$\xi \simeq 1 - p_0$: initial state fidelity
λ: 1-norm
C_{gs}: Trotter constant

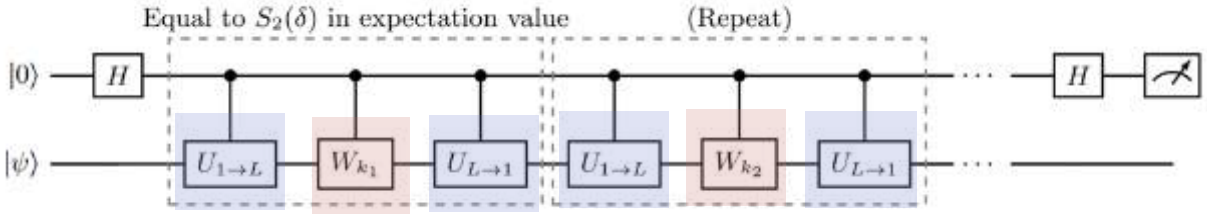
Method	Maximum per-shot simulation cost	Total simulation cost
Deterministic (p-th order Suzuki-Trotter)	$\mathcal{O}(L\xi C_{\mathrm{gs}}^{1/p}\epsilon^{-1-1/p})$	$\mathcal{O}(L\xi^{-1}C_{\mathrm{gs}}^{1/p}\epsilon^{-1-1/p})$
Randomized	$\mathcal{O}(\lambda^2\xi^2\epsilon^{-2})$	$\mathcal{O}(\lambda^2\epsilon^{-2})$
Partially randomized	$\mathcal{O}(L_D\xi C_{\mathrm{gs}}^{1/p}\epsilon^{-1-1/p} + \lambda_R^2\xi^2\epsilon^{-2})$	$\mathcal{O}(L_D\xi^{-1}C_{\mathrm{gs}}^{1/p}\epsilon^{-1-1/p} + \lambda_R^2\epsilon^{-2})$

$$\hat{H} = \underbrace{\sum_{\ell=1}^{L_D} \hat{H}_\ell}_{\text{Deterministic}} + \underbrace{\sum_{\ell=L_D+1}^{L} \hat{H}_\ell}_{\text{Randomized}},$$

✓ Conditions under which partial randomization reduces the computational cost.

$$\lambda_R \ll \lambda \quad \text{and} \quad L_D \ll L$$

$$\lambda_R := \sum_{\ell=L_D+1}^{L} |c_\ell|.$$



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Research idea

- Cost of partially randomized Trotterization

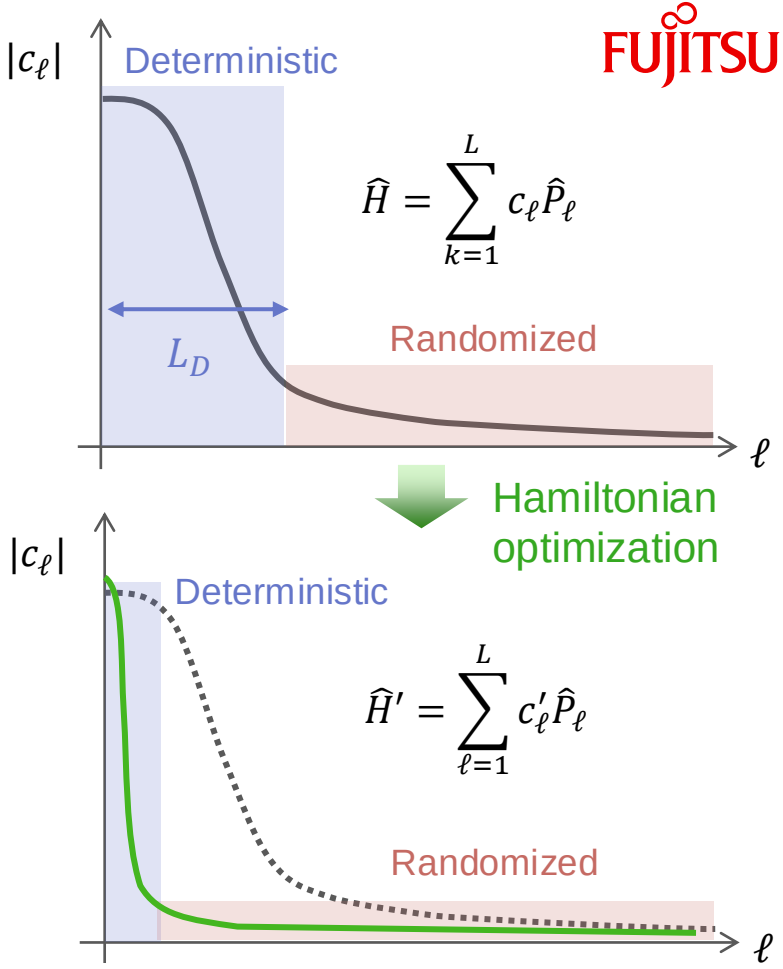
$$G = \min_{L_D} \left[\alpha L_D + \beta \lambda_R^2 \right] \qquad \lambda_R := \sum_{\ell=L_D+1}^{L} |c_\ell|.$$

L_D: Number of deterministic terms
λ_R: Total weight of the randomized terms
α, β: Algorithm-dependent constants, independent of the coefficient distribution

Conditions for substantial cost reduction

① The weight is concentrated in a few dominant terms (L_D is small)
② The tail terms have a small total weight (λ_R is small)

Develop a Hamiltonian optimization method that produces such a favorable distribution.



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Molecular Hamiltonian optimization

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- Electronic structure Hamiltonian

$$\hat{H} = \sum_{pq}^{N} \sum_{\sigma=\uparrow,\downarrow} h_{pq} \hat{a}^{\dagger}_{p\sigma} \hat{a}_{q\sigma} + \frac{1}{2} \sum_{pqrs}^{N} \sum_{\sigma\tau} g_{pqrs} \hat{a}^{\dagger}_{p\sigma} \hat{a}^{\dagger}_{r\tau} \hat{a}_{s\tau} \hat{a}_{q\sigma}$$

Majorana representation

$$\hat{\gamma}_{p\sigma} := \hat{a}_{p\sigma} + \hat{a}^{\dagger}_{p\sigma}, \; \hat{\tilde{\gamma}}_{p\sigma} := -i(\hat{a}_{p\sigma} - \hat{a}^{\dagger}_{p\sigma})$$

$$\hat{H} = \frac{i}{2} \sum_{pq,\sigma} \left(h_{pq} + \sum_{r} g_{pqrr} - \frac{1}{2} \sum_{r} g_{prrq} \right) \hat{\gamma}_{p\sigma} \hat{\tilde{\gamma}}_{q\sigma}$$
$$+ \frac{1}{4} \sum_{p>r,s>q} \sum_{\sigma} (g_{pqrs} - g_{psrq}) \hat{\gamma}_{p\sigma} \hat{\gamma}_{r\sigma} \hat{\tilde{\gamma}}_{q\sigma} \hat{\tilde{\gamma}}_{s\sigma} + \frac{1}{8} \sum_{pqrs} \sum_{\sigma \neq \tau} g_{pqrs} \hat{\gamma}_{p\sigma} \hat{\gamma}_{r\tau} \hat{\tilde{\gamma}}_{q\sigma} \hat{\tilde{\gamma}}_{s\tau}.$$

- Qubit Hamiltonian (Pauli LCU)

$$\hat{H} = \sum_{\ell=1}^{L} c_{\ell} \hat{P}_{\ell},$$

Equivalent

- Optimize the electronic integrals h_{pq} and g_{pqrs} in the Majorana representation to reduce cost metrics such as the 1-norm.
- Perform the optimization using transformations that preserve the relevant spectrum, and hence the target energy eigenvalues.

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Molecular Hamiltonian optimization

- **Orbital Optimization (OO)** *

*Koridon et al., Phys. Rev. Res. 3, 033127 (2021).

  - Rotate the molecular-orbital basis used to represent the Hamiltonian.

$$\hat{a}'_{i\sigma} = \sum_{p}^{N} (U_{\mathrm{OO}})_{ip} \hat{a}_{p\sigma}, \quad U_{\mathrm{OO}} := e^{-K}$$

$$K := \begin{pmatrix} 0 & \kappa_{12} & \kappa_{13} & \cdots & \kappa_{1N} \\ -\kappa_{12} & 0 & \kappa_{23} & \cdots & \kappa_{2N} \\ -\kappa_{13} & -\kappa_{23} & 0 & \cdots & \kappa_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -\kappa_{1N} & -\kappa_{2N} & -\kappa_{3N} & \cdots & 0 \end{pmatrix}$$

  - Because this is a unitary transformation, the spectrum of H remains unchanged.

$$\hat{H}_{\mathrm{OO}}(\boldsymbol{\kappa}) = \sum_{pq}^{N} \sum_{\sigma=\uparrow,\downarrow} \tilde{h}_{pq}(\boldsymbol{\kappa}) \hat{a}^{\dagger}_{p\sigma} \hat{a}_{q\sigma} + \frac{1}{2} \sum_{pqrs}^{N} \sum_{\sigma,\tau=\uparrow,\downarrow} \tilde{g}_{pqrs}(\boldsymbol{\kappa}) \hat{a}^{\dagger}_{p\sigma} \hat{a}^{\dagger}_{r\tau} \hat{a}_{s\tau} \hat{a}_{q\sigma}$$

of parameters: $N(N-1)/2$

$$\tilde{h}_{pq}(\boldsymbol{\kappa}) = \sum_{ij}^{N} [\exp(-\boldsymbol{\kappa})]_{ip} h_{ij} [\exp(-\boldsymbol{\kappa})]_{jq},$$

$$\tilde{g}_{pqrs}(\boldsymbol{\kappa}) = \sum_{ijkl}^{N} [\exp(-\boldsymbol{\kappa})]_{ip} [\exp(-\boldsymbol{\kappa})]_{jq} g_{ijkl} [\exp(-\boldsymbol{\kappa})]_{kr} [\exp(-\boldsymbol{\kappa})]_{ls}$$

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Molecular Hamiltonian optimization

- **Block-Invariant Symmetry Shift (BLISS) ***

  * Loaiza & Izmaylov, JCTC., 19, 8201 (2023).

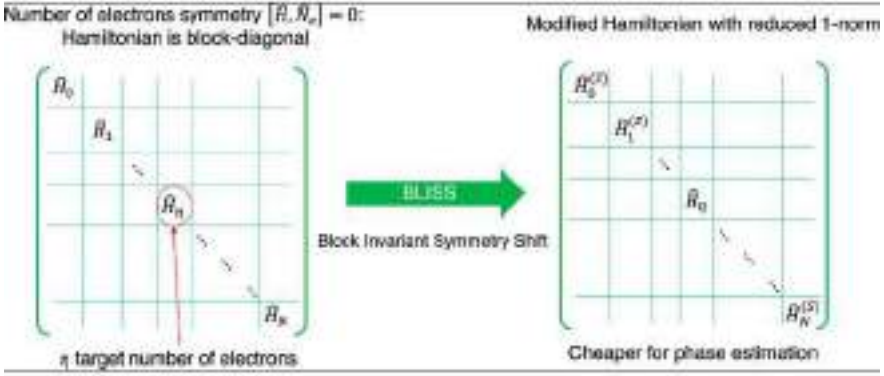
  - Apply a shift transformation based on symmetries of the electronic Hamiltonian, such as particle number and spin.
    - Subtract a shift operator chosen to leave the spectrum of the target symmetry sector unchanged.

$$\hat{H}_{\mathrm{BLISS}}(\boldsymbol{\tau},\boldsymbol{\xi}) = \hat{H} - \hat{K}(\boldsymbol{\tau},\boldsymbol{\xi}) = \sum_{pq}^{N}\sum_{\sigma=\uparrow,\downarrow} \tilde{h}_{pq}(\boldsymbol{\tau},\boldsymbol{\xi})\hat{a}^{\dagger}_{p\sigma}\hat{a}_{q\sigma} + \frac{1}{2}\sum_{pqrs}^{N}\sum_{\sigma,\tau=\uparrow,\downarrow} \tilde{g}_{pqrs}(\boldsymbol{\tau},\boldsymbol{\xi})\hat{a}^{\dagger}_{p\sigma}\hat{a}^{\dagger}_{r\tau}\hat{a}_{s\tau}\hat{a}_{q\sigma}$$

$$\hat{K}(\boldsymbol{\tau},\boldsymbol{\xi}) = \tau_1\left(\hat{N}_e - N_e\right) + \tau_2\left(\hat{N}_e^2 - N_e^2\right) + \sum_{pq}\xi_{pq}\hat{F}_{pq}\left(\hat{N}_e - N_e\right)$$

$$\hat{N}_e = \sum_{p\sigma}\hat{a}^{\dagger}_{p\sigma}\hat{a}_{p\sigma} \qquad \hat{F}_{pq} = \sum_{\sigma}\hat{a}^{\dagger}_{p\sigma}\hat{a}_{q\sigma}$$

The shift operator vanishes in the N_e-electron sector.
→ **The spectrum in the target subspace remains unchanged.**



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Unitary Weight Concentration (UWC)

- Optimize the distribution of the Hamiltonian coefficients to minimize the gate count of partially randomized Trotterization.
  - Hamiltonian（parametrized by OO and BLISS）

$$\hat{H}'(\vec{\chi}) = \sum_{\ell} c'_{\ell}(\vec{\chi})\hat{P}_{\ell}.$$

☜ $\vec{\chi}$ contains $O(N^2)$ variational parameters

Step function

$$\Theta(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

  - Objective function: maximum gate count for RPE

$$G(\vec{\chi}) = \min_{L_D} \left[\alpha L_D + \beta \lambda_R^2 \right] = \min_{L_D} \left[\alpha \sum_{\ell=1}^{L} \Theta(L_D - \ell) + \beta \left(\sum_{\ell=1}^{L} |c'_{\ell}(\vec{\chi})| \cdot [1 - \Theta(L_D - \ell)] \right)^2 \right]$$

    - **Nested optimization**: Evaluate G requires minimization over L_D.
    - **Non-differentiability**: Determining the optimal L_D requires sorting the coefficients $\{c'_{\ell}\}$, resulting in discontinuities.

⊗ Direct optimization of the gate count is numerically challenging.

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Unitary Weight Concentration (UWC)

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- Optimize the distribution of the Hamiltonian coefficients to minimize the gate count of partially randomized Trotterization.
  - Objective function: maximum gate count for RPE

$$G(\vec{\chi}) = \min_{L_D} \left[\alpha L_D + \beta \lambda_R^2 \right] = \min_{L_D} \left[\alpha \sum_{\ell=1}^{L} \Theta(L_D - \ell) + \beta \left(\sum_{\ell=1}^{L} |c'_\ell(\vec{\chi})| \cdot [1 - \Theta(L_D - \ell)] \right)^2 \right]$$

✓ Smooth the step function Θ using sigmoid function $\varsigma(x) = \frac{1}{1+e^{-x}}$.

  - Surrogate model

$$\widetilde{G}(\vec{\chi}) = \alpha \sum_{\ell=1}^{L} \varsigma \left(\frac{|c'_\ell(\vec{\chi})| - \tau}{\epsilon_{\text{reg}}} \right) + \beta \left(\sum_{\ell=1}^{L} |c'_\ell(\vec{\chi})| \cdot \left[1 - \varsigma \left(\frac{|c'_\ell(\vec{\chi})| - \tau}{\epsilon_{\text{reg}}} \right) \right] \right)^2$$

    - $\tau = (|c_{L_D}| + |c_{L_D+1}|)/2$: Midpoint between the coefficients adjacent to the optimal partition L_D for the initial distribution
    - ϵ_{reg}: Hyperparameter controlling the sigmoid smoothness ($\epsilon_{\text{reg}} \to 0$ recovers the step function)

✓ Differentiable surrogate objective（avoid nested optimization and discontinuities）

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